Amazon Using Machine Learning

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Amazon + Machine Learning

- Amazon founded: 1994
- Personalized recommendations: 1998
- ML recommendation technology: 2003
- **Today:** ML is used across sales, marketing, inventory and logistics.

Amazon - Use of Machine Learning

Amazon is one of the world's largest e-commerce companies.

Amazon uses Machine Learning (ML) to understand customer behavior.

ML is used for:

- Product recommendations
- Demand forecasting
- Personalized advertising
- Fraud detection
- Inventory management

Main goal: Improve customer experience and increase business revenue.

How ML Works in Amazon

Amazon Recommendation System

Customer Data

↓

Search History
Products Viewed
Previous Purchases
Cart Activity

↓

Machine Learning

↓

Understand Customer Preference

↓

Recommend Products

↓

Customer Purchase

Amazon Recommendation System

Example

If a customer frequently searches for mobile phones, Amazon's ML system can recommend:

- 📱 Mobile phones
- 🎧 Earphones
- 🔌 Chargers
- 📱 Mobile covers

Result: The customer sees products that are more relevant to them.

Challenges of Using ML in Amazon

1. **Huge Amount of Data**
   Amazon has millions of customers and products, so processing data is complex.
2. **Changing Customer Preferences**
   Customer interests change over time.
3. **New Customers**
   A new customer has little purchase history, making recommendations difficult.
4. **New Products**
   A newly launched product has little customer data.
5. **Privacy & Security**
   Customer data must be protected and used responsibly.
6. **Accuracy**
   Wrong recommendations can reduce customer trust and sales.

Impact on Sales

Business Impact of ML

ML helps Amazon achieve:

↑ Product Sales

Relevant recommendations can encourage additional purchases.

↑ Conversion Rate

Customers are more likely to purchase products relevant to their interests.

↑ Average Order Value

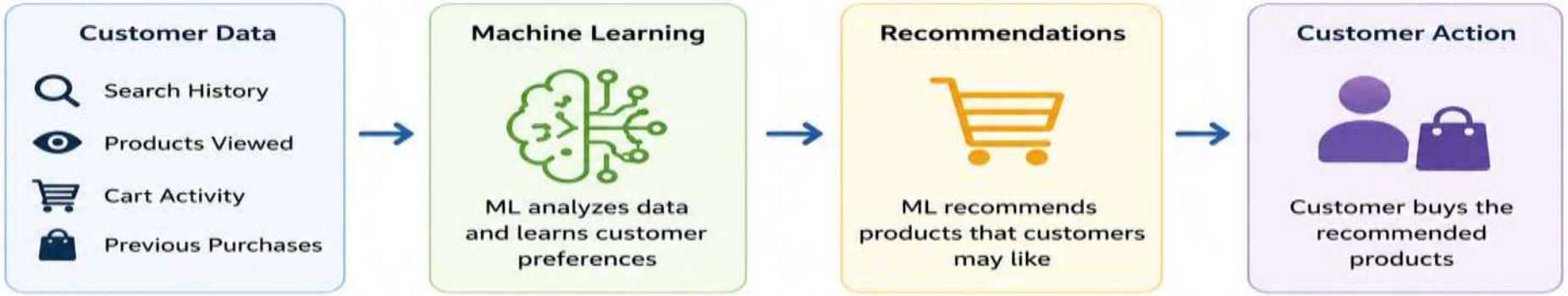
Recommendations can encourage customers to buy additional products.

↓ Inventory Problems

Demand forecasting helps Amazon plan inventory.

Amazon Uses Machine Learning

amazon



Impact on Business



Thank You